SWE363 · Assignment 4

Mashael Alghamdi

Personal Portfolio Website

HTML · CSS · JavaScript | Wikipedia REST API

Project Overview

What is it?

A fully static personal portfolio website showcasing projects, skills, and academic journey — built with pure HTML, CSS, and JavaScript with no frameworks.

Objectives

- Demonstrate advanced front-end skills
- Integrate a live external API
- Apply state management & performance optimization
- Deliver a polished, responsive, deployed web app

Personal Motivation

- Build something I can actually keep and share
- Put all course concepts into one real project
- Practice clean, maintainable code habits
- Create a portfolio ready for internship applications

TECHNICAL DEMONSTRATION

Key Features

🔍 Search & Sort

Real-time filtering by first word + A→Z / Z→A sorting with data-index reset

📂 Accordion

Expand one project at a time using max-height animation — no layout shift

🌐 Wikipedia API

Fetch programming facts with retry logic (10 attempts) and duplicate prevention via Set

🌙 Dark / Light Mode

Theme toggle persisted in localStorage — restored instantly on page load

⏱️ Visit Timer

Counts seconds live, switches to Xm Xs format after 60s

✅ Form Validation

Regex email check, .trim() whitespace guard, custom validity messages + fade feedback

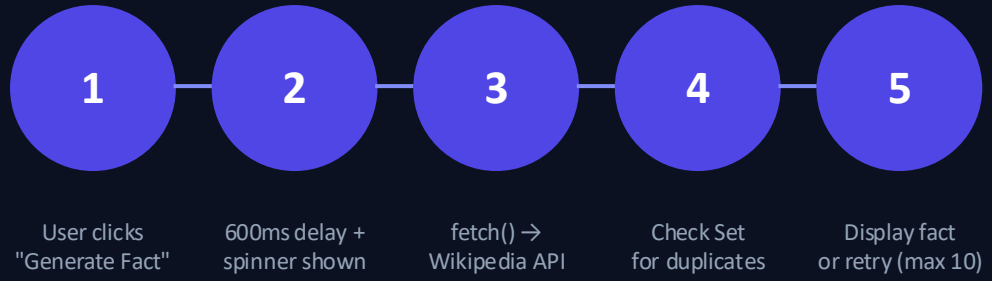
TECHNICAL DEMONSTRATION

Architecture & API Flow

File Structure

```
Assignment4/
  index.html
  css/
    styles.css
  js/
    script.js
  assets/images/
  docs/
    ai-usage-report.md
    technical-doc.md
```

Wikipedia API Flow



TECHNICAL DEMONSTRATION

AI Integration

Claude

Analysis & Debugging

- Designed Wikipedia API retry logic
- Improved form validation (.trim, regex)
- Fixed layout and accordion behavior

ChatGPT

Concepts & New Features

- Explained search filtering & event handling
- Assisted in building the Journey timeline section
- Assisted with scroll animation logic
- Improved documentation structure

GitHub Copilot

Code Completion

- Real-time suggestions in VS Code
- Reduced syntax errors

TECHNICAL DEEP DIVE

Challenges & Lessons

Resolved

- API duplicate facts via Set + retry loop
- Form whitespace bypass using .trim()
- Accordion closing others on open
- Sort order reset with data-index
- Journey section layout alignment

Unresolved

- Contact form doesn't send real emails (no backend)
- Wikipedia API can fail with unexpected data
- Timeline uses scroll event (less efficient than IntersectionObserver)

Planned

- Backend integration for real contact form
- More projects & filtering by category

CONCLUSION

Outcomes & Next Steps

What I Achieved

- Fully deployed portfolio at mashaelalghamdi.github.io
- 10 JavaScript features from scratch — no libraries
- Live API integration with retry & deduplication logic
- Responsive design working on mobile and desktop
- Clean, documented, version-controlled codebase

Future Improvements

- Real backend for contact form (Node.js / EmailJS)
- Project category filter tags
- Migrate timeline to IntersectionObserver
- Add more projects over time
- Expand to include a blog section